

Abstract Algebra III: Assignment - Week 14

钟皓宇(12533556)

May 27, 2026

Problem 1

Let G be a finite group and let p be a prime. Prove that each element $x \in G$ can be uniquely written as

$$x = st,$$

where s is p -regular, t is p -singular, and $st = ts$. If $x_1 = s_1t_1$ is such a decomposition of an element x_1 that is conjugate to x , then s is conjugate to s_1 , and t is conjugate to t_1 .

Solution: Let $x \in G$, and write $|x| = n = p^am$, where $(m, p) = 1$. Choose integers A, B such that $Ap^a + Bm = 1$. Set $s = x^{Ap^a}$ and $t = x^{Bm}$. Then s and t commute, since both are powers of x , and $st = x^{Ap^a+Bm} = x$. Moreover, $s^m = x^{Ap^am} = x^{An} = 1$, so $|s| \mid m$; hence s is p -regular. Similarly, $t^{p^a} = x^{Bmp^a} = x^{Bn} = 1$, so $|t| \mid p^a$; hence t is a p -element. Thus the required decomposition exists.

Now suppose $x = s_1t_1$ is another such decomposition, with s_1 p -regular, t_1 a p -element, and $s_1t_1 = t_1s_1$. Since s_1 and t_1 commute and have relatively prime orders, $|x| = |s_1t_1| = |s_1||t_1|$. Comparing the p -part and the prime-to- p part of $|x| = p^am$, we get $|s_1| = m$ and $|t_1| = p^a$. Therefore

$$x^{Ap^a} = (s_1t_1)^{Ap^a} = s_1^{Ap^a}t_1^{Ap^a} = s_1,$$

because $Ap^a \equiv 1 \pmod{m}$ and $t_1^{p^a} = 1$. Similarly, $x^{Bm} = (s_1t_1)^{Bm} = t_1$. Hence $s_1 = s$ and $t_1 = t$, so the decomposition is unique.

Finally, if $x_1 = gsg^{-1}$, then $x_1 = (gsg^{-1})(gtg^{-1})$. The first factor is still p -regular, the second is still a p -element, and they commute. By uniqueness, if $x_1 = s_1t_1$ is the corresponding decomposition, then $s_1 = gsg^{-1}$ and $t_1 = gtg^{-1}$. Thus s is conjugate to s_1 , and t is conjugate to t_1 . ■